Research Paper 1: Sustainable Energy Management in the AI Era: A Comprehensive Analysis of ML and DL Approaches

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Abstract

This paper presents a comprehensive analysis of sustainable energy management using Machine Learning (ML) and Deep Learning (DL) techniques. The main idea is to use AI-based methods to improve energy consumption, predict electricity demand, and optimize energy resources. The paper reviews more than 200 research studies to understand the applications of ML and DL in load forecasting, demand response, and renewable energy management. The study highlights the benefits of these techniques in improving prediction accuracy, energy efficiency, and power grid stability. It also discusses challenges such as limited real-world implementation, data quality issues, and model interpretability.

Introduction

Increasing energy demand and the need for sustainable energy utilization make efficient energy management important. Traditional energy management systems have limitations in accurately predicting electricity demand and optimizing energy consumption. Machine Learning (ML) and Deep Learning (DL) can analyse energy data and improve forecasting and decision-making. This paper reviews the applications of these techniques in load forecasting, demand response, and renewable energy management to support efficient and sustainable energy systems.

Proposed System

The proposed system uses Machine Learning (ML) and Deep Learning (DL) techniques to predict energy demand and improve energy management. Energy data is collected and analysed to identify consumption patterns and optimize electricity usage. AI-based models support demand forecasting, renewable energy integration, and efficient resource utilization. This system helps improve energy efficiency and power grid stability.

Working

The system collects energy-consumption data and processes it using Machine Learning (ML) and Deep Learning (DL) algorithms. These algorithms analyse historical data to predict future electricity demand and identify consumption patterns. The system supports demand-response strategies and renewable energy management to optimize energy usage. The results help improve energy efficiency, reduce wastage, and maintain power grid stability.

Case Study and Findings

The paper reviews more than 200 research studies on the use of Machine Learning (ML) and Deep Learning (DL) in sustainable energy management. The findings show that these techniques improve electricity demand forecasting, optimize energy consumption, and support renewable energy integration. They also help improve energy efficiency and power grid stability. However, challenges such as limited real-world implementation, data quality, and model interpretability still need to be addressed.

Advantages

- Improves electricity demand forecasting accuracy.
- Increases energy efficiency and reduces energy wastage.
- Optimizes energy consumption and resource utilization.
- Supports the integration of renewable energy sources.
- Improves power grid stability and reliability.
- Enables better energy-management decisions using AI.

Limitations

The system depends on accurate energy data and reliable ML/DL models. Poor data quality and limited real-world implementation can affect its performance. Understanding how complex AI models make predictions can also be difficult. Further research is needed to overcome these challenges.

Conclusion

The paper demonstrates that Machine Learning (ML) and Deep Learning (DL) can make energy management more efficient and sustainable. By predicting electricity demand and optimizing energy consumption, these techniques can reduce energy wastage and support renewable energy integration. This concept is useful for a Smart Power Management project using sensors, a microcontroller, and a web or mobile interface.

Note: This is an original student-friendly summary of the cited research paper, prepared for SPI academic submission; it is not a reproduction of the original paper.

Research Paper 2: Artificial Intelligence, IoT, and Solar PV-Integrated Home Energy Management Systems: A Review

Authors/Source: Md. Rokonuzzaman et al.

Original publication: Results in Engineering, Volume 29, Article 108954, 2026. DOI: 10.1016/j.rineng.2025.108954

Abstract

This paper reviews the use of Artificial Intelligence (AI) and the Internet of Things (IoT) in Home Energy Management Systems (HEMS). It focuses on energy forecasting, appliance scheduling, solar energy integration, and demand-side management. The study highlights how AI-based techniques can improve energy efficiency and reduce electricity costs. It also discusses challenges such as data security, computational complexity, and limited real-world implementation.

Introduction

Increasing energy demand and electricity costs make efficient home energy management important. Traditional systems have limitations in optimizing energy consumption and managing renewable energy sources. AI and IoT technologies can help monitor energy usage and improve decision-making. This paper reviews smart home energy management systems integrated with solar photovoltaic (PV) technology.

Proposed System

The paper reviews an integrated Home Energy Management System that combines AI, IoT, solar PV panels, and energy storage systems. Smart devices collect energy data from household appliances and communicate with the management system. AI algorithms analyse the data to optimize appliance scheduling and energy consumption. This approach supports efficient energy usage and better integration of renewable energy.

Working

The system collects household energy-consumption data using IoT-enabled devices. AI algorithms analyse the data to forecast electricity demand and determine suitable appliance schedules. Solar PV generation and stored energy are considered to improve energy utilization. The system aims to reduce electricity costs, minimize energy wastage, and improve overall efficiency.

Case Study and Findings

The paper reviews existing research on AI- and IoT-integrated home energy management systems. The findings show that advanced AI models can improve energy forecasting, appliance scheduling, and demand-side management. Reinforcement learning also offers potential for adaptive, real-time energy control. However, challenges such as communication standards, cybersecurity, and insufficient real-world testing remain.

Advantages

• Improves household energy efficiency.
• Helps reduce electricity costs.
• Supports solar energy integration.
• Optimizes appliance scheduling.
• Improves electricity demand forecasting.
• Supports intelligent and automated energy management.

Limitations

The system may require additional hardware, computing resources, and communication infrastructure. Different IoT devices may face compatibility issues due to fragmented communication standards. Cybersecurity vulnerabilities and data privacy concerns can affect system reliability. Limited real-world testing also creates challenges for large-scale implementation.

Conclusion

The paper demonstrates that AI and IoT can improve home energy management by optimizing electricity consumption and integrating solar energy. Intelligent forecasting and appliance scheduling can help reduce energy costs and improve efficiency. This concept is useful for a Smart Power Management project using sensors, a microcontroller, solar energy monitoring, and a web or mobile interface.

Note: This is an original student-friendly summary of the cited research paper, prepared for SPI academic submission; it is not a reproduction of the original paper.

Research Paper 3: Smart Energy Management System Using Machine Learning

Authors/Source: Ali Sheraz Akram, Sagheer Abbas, Muhammad Adnan Khan, Atifa Athar, Taher M. Ghazal, and Hussam Al Hamadi.

Original publication: Computers, Materials & Continua, Volume 78, Issue 1, 2024. DOI: 10.32604/cmc.2023.032216

Abstract

The research proposes an Intelligent Energy Management System (IEMS) for smart cities using Machine Learning (ML). It uses the Internet of Energy (IoE) network to collect energy-related data and applies ML to improve energy management. The system aims to reduce energy consumption, minimize electricity costs, improve efficiency, and reduce carbon emissions. The proposed model achieved 92.11% accuracy and a 7.89% miss rate during validation.

Introduction

Increasing energy consumption has become a major challenge due to modernization and the growing use of electrical appliances. Excessive energy consumption increases electricity costs, depletes natural resources, and contributes to environmental pollution. Traditional energy management systems face difficulties in balancing energy consumption and user comfort. Therefore, the paper proposes an intelligent system that uses IoT and Machine Learning to improve energy efficiency in smart cities.

Proposed System

The proposed system uses an Internet of Energy (IoE) network and a Support Vector Machine (SVM) algorithm for intelligent energy management. It consists of the following layers:

- Sensory Layer: Collects energy-related data.
- Pre-processing Layer: Handles missing and noisy data using moving averages and normalization.
- Training Layer: Trains the SVM model using collected data.
- Performance Layer: Evaluates the model's accuracy and miss rate.
- Cloud Storage: Stores processed data for future prediction and validation.

Working

The system collects energy-management data through the IoE network and sends it to the preprocessing layer. The cleaned data is divided into 70% training data and 30% testing data. The SVM algorithm learns patterns from the training data and predicts whether energy management is required. The system evaluates the prediction results and stores suitable outputs in cloud storage for future use.

Case Study and Findings

The proposed system was evaluated using a dataset of 757 records, divided into 529 training samples and 228 validation samples. The results showed:

- Training accuracy: 93.38%
- Validation accuracy: 92.11%
- Validation sensitivity: 92.71%
- Validation specificity: 91.67%
- Validation miss rate: 7.89%

The proposed model achieved better accuracy than the existing approaches compared in the paper, including linear regression, polynomial regression, decision-tree regression, and fused machine learning.

Advantages

• Improves energy-management prediction using Machine Learning.
• Reduces unnecessary energy consumption and associated costs.
• Uses preprocessing to handle missing and noisy data.
• Supports intelligent energy management in smart cities.
• Helps reduce carbon emissions and improve energy efficiency.
• Uses cloud storage for data storage and validation.

Limitations

The system depends on the quality and availability of energy data, reliable communication infrastructure, and cloud storage. Implementing Machine Learning requires computational resources and increases system complexity. Furthermore, the proposed model was evaluated using a dataset and simulation results, so further real-world testing would help verify its performance in practical applications.

Conclusion

The paper presents an Intelligent Energy Management System using Machine Learning and the Internet of Energy. The SVM-based model achieved 92.11% validation accuracy and a 7.89% miss rate, outperforming the previous approaches compared in the study. The system demonstrates the potential of ML for improving energy efficiency, reducing energy costs, and supporting sustainable smart cities.

Note: This is an original student-friendly summary of the cited research paper, prepared for SPI academic submission; it is not a reproduction of the original paper.